

Computer vision

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Course: Computer vision

Course code: 1TA0506

Ans:

a) Computer vision is a branch of Artificial intelligence.

- The Computer vision give the tasks to computer to create a Photos,

- sampling and quantization image foundation models.

b) ★ The Two application of the Computer vision is image processing and the Computer vision. quantization and the samplings.

2)

Ans:

a) The feature extraction is the process of the identifying important characteristics such as the edges, corners, of the computer visions.

b) The object Recognition in the computer vision it identifies and classifying the object accurately.

3)

Ans

a) i) The low level images processings enhances and noise removal.

ii) The mid level images process improve the features.

b) . The mid level of image processing features and segments the image.

. The high level understand the image and recognize object.

Ans:

a) The pixels is the smallest unit of digital images.

b) Higher resolution gives cleaner and more detailed images.

• lower resolution reduces image quality.

5)

Ans

a) • IE captures light from real world scene.
• The sensor convert light into electrical signals.

x

b)

i) Digital signals

ii) camera

i) Sampling converts a continuous image into pixels.

ii) Together they create a digital image.

iii) Sampling is the process of dividing an image into pixels.

b) It converts pixel value into discrete intensity levels.

3) Ans:

a) JIE Explains how light travel from an object through a camera.

ii) It projects a 3D object onto a 2D image plane.

b) illumination provides light, making objects visible and affecting image quality.

Ans:

a) i) image processing improves images.

ii) computer vision understand and interprets images.

iii) computer vision use image processing as a preprocessing step.

b) i) image processing enhances images, computer vision interprets them.

ii) computer vision output decision.

g)

a) i) used in healthcare, autonomous vehicles.

ii) detect diseases from X-rays, CT scans

b)

i) detects roads, pedestrians, traffic signals, vehicles for safe driving.

10)

a)

i) Digital images are represented as pixels.

ii) pixels are basic building blocks of vision accuracy.

b)

i) The image resolution influence image analysis of the computer vision.